

第五十五屆全國技能競賽 青少年組商務軟體設計職類 競賽試題暨評分彙總表



競賽試題暨評分項目

☒ 商用軟體設計一 Software Design 1	☐ 商用軟體設計二 Software Design 2
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選 手 編 號：_____

崗 位 編 號：_____

選 手 姓 名：_____

總 分：_____

評 分 老 師：_____

1. Scenario Description

The contestant is employed at a multinational software design consulting firm and has recently undertaken a project commissioned by **Zyberion**.

Zyberion is a leading electric vehicle (EV) manufacturer in China, widely favored by consumers for its high cost-performance ratio, advanced technology, and innovative features. The brand focuses on technological innovation, offering high-performance, environmentally friendly smart mobility solutions. Its product lineup includes fully electric sedans and electric SUVs, with ongoing developments in areas such as autonomous driving, smart cockpits, and vehicle-to-everything (V2X) communication.

To respond to rapid market changes, Zyberion is currently implementing a “**ZCharge Plan**” — a proprietary EV charging network initiative — and has commissioned your company to design the management system and underlying database for this plan. In addition, Zyberion established **Zyberion Taiwan** in 2021 to provide premium electric vehicle services in the Taiwan market. Your company has also been tasked with developing both the management platform and a client-facing mobile application called “**Zyberion Taiwan**.”

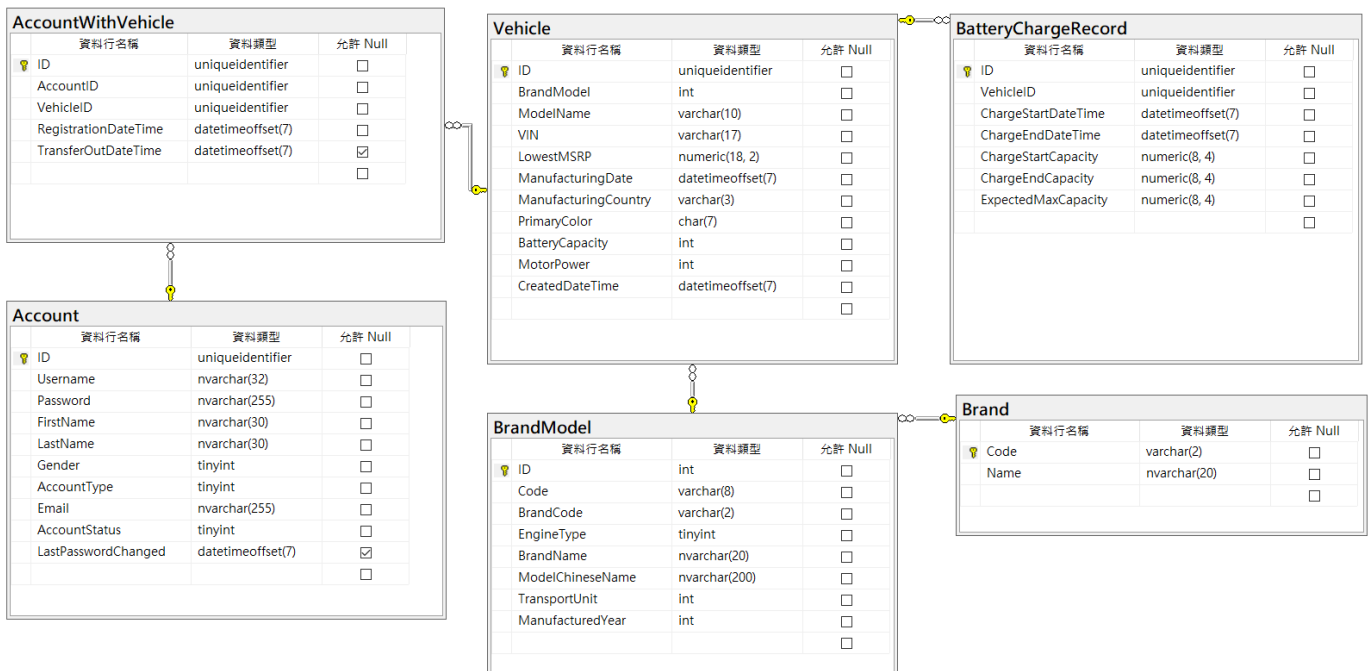
You are currently stationed at Zyberion’s EV manufacturing plant in **Shanghai, China**, serving as an IT systems analyst. In response to the fast-evolving EV market and rapidly advancing technologies, Zyberion places high importance on its IT personnel. The company is actively planning to develop its proprietary charging network system, named “**ZCharge Plan**.”

You have been assigned as the **Information Systems Analyst** for this system development project. Your responsibilities include systems analysis, database design, and the creation of relevant diagrams and documentation to communicate the system's functionalities to internal management and potential investors.

2. Database Description

Your project manager has already completed the system analysis and preliminary SQL database design for Subject One. The backup database file is named **Zyberion_ANM.bak**. You are required to use the provided SQL database to further develop and complete the system functionalities. Some additional system features may require you to expand the database by adding new tables and columns.

1. Please rename the SQL backup database file **Zyberion_ANM.bak** by replacing the last two characters “**NM**” with your assigned station number.
2. The database entity-relationship diagram (ERD) is shown below:



3. Guidelines

- Certain system functionalities require you to **add new fields**. All added fields must begin with the prefix **New_** (e.g., **New_MyColumn1**) and **must be named in English**. Fields that do not follow this naming convention will **not be scored**.
- The **solution file name** must be **ZCP.sln**. System functionalities must be implemented according to the specifications provided in the "Design Requirements" section. **1 point will be deducted for incorrect file names**.
- The **database name** linked to the application must be **Zyberion_ANM**, where the last two characters **NM** should be replaced with your assigned station number. **1 point will be deducted for incorrect database names**.
- When submitting the **SQL file**, the backup file must be named **Zyberion_ANM.bak** (**NM** should be replaced with your station number). **1 point will be deducted for incorrect file names or extensions**. If the database cannot be restored, 20 points will be deducted from the total score.
- When submitting **source code files**, failure to correctly or completely upload files to the **FTP server** will result in a **20-point deduction** from the total score.

- A ReadMe.txt file must be placed in the **project directory** (or the output directory of the application) to describe the **login accounts and passwords for users with different roles**. Failure to provide this file or incorrect file naming will result in a **2-point deduction**.
- The **User Interface** must comply with the design specifications provided:
 - Missing "**System Name**", "**Brand Identity (Logo)**", "**System Time**", or "**Current User Name**" will result in **1-point deduction each**.
 - Incorrect display of the current user name format (must be: {First Name}{single space}{Last Name}) will result in a **1-point deduction**.
 - The **Login/Logout** button must accurately reflect the current login status. (**1 point**)
 - Failure to use the **specified color palette, brand identity, or fonts** according to the guidelines will result in a **1-point deduction each**.

4. Vehicle Models Description

ZYBERION TAIWAN offers two main vehicle models:

- **ZYBERION ZEV1** – A four-seater luxury sedan
- **ZYBERION ZEVX** – A seven-seater luxury SUV

The specifications of these models are outlined in the table below:



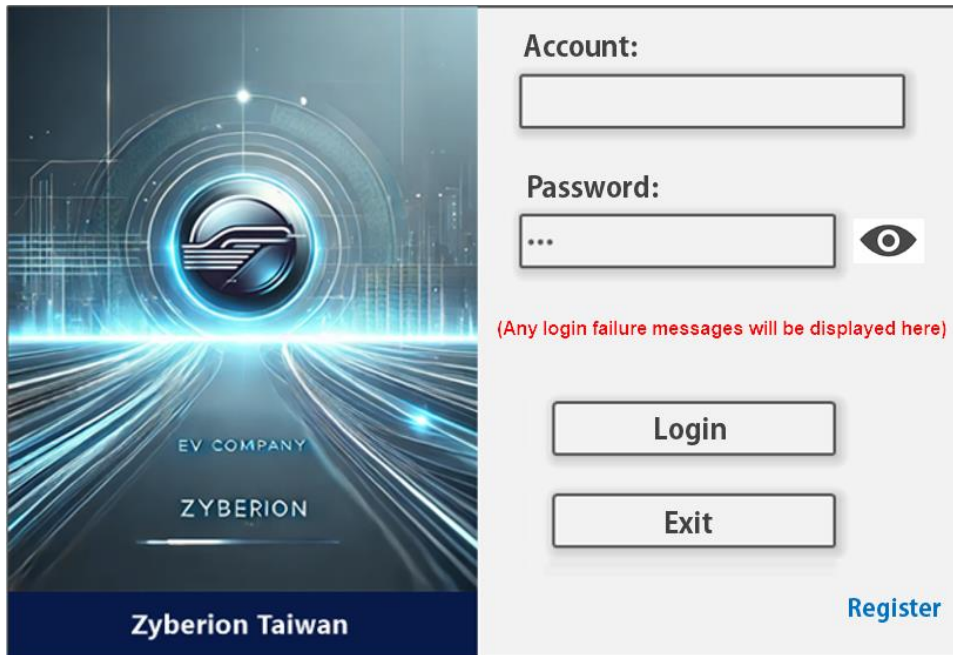
Brand	ZYBERION TAIWAN	ZYBERION TAIWAN
Model	ZEV1	ZEVX
Origin / Country of Manufacture	US / TW	
Color Options / Color Scheme	Tech Blue/ Trendy Brown	
Year	2021~2024	
Seating Capacity	4	7
Horsepower	170 hp	204 hp
Suggested Retail Price (USD)	48,500	59,500
Battery Capacity	63 kWh	82 kWh
Motor Power Output	125 kW	150 kW
Charging Time	Fast charging: charges up to 80% in 50 minutes. Standard charging: fully charges in approximately 9 hours.	

5. Design Requirements

The proprietary charging network system (ZCP) management platform for Zyberion has two main functions: **Login and Register Accounts**, and **Vehicle query**. The detailed descriptions of each function are as follows:

(1) Login and Register

Login & Logout



The image shows a login interface for 'Zyberion Taiwan'. On the left is a blue-themed graphic with a circular logo and the text 'EV COMPANY' and 'ZYBERION'. On the right is a white login form. The form has an 'Account:' label above a text input field, a 'Password:' label above a masked password input field (indicated by three dots), and an eye icon to toggle visibility. Below the password field is a red text placeholder: '(Any login failure messages will be displayed here)'. At the bottom of the form are 'Login' and 'Exit' buttons. A 'Register' link is located at the bottom right of the form area.

1. Provide an admin-only login interface. Users can log in using their **Username** and **Password**. The password input box should mask the characters by default. When the mouse hovers over the "👁" icon, the password becomes visible; when the mouse moves away, it returns to masked characters.
2. If either the username or password is incorrect, display the message:
"Incorrect username or password, error count: n times."
3. If incorrect login attempts exceed 3 times (exclusive), the account is set to **disabled**.
4. If the credentials are correct but the account is disabled, display:
"Account has been locked and cannot log in."
Login is not allowed.

The account status (Account.AccountStatus) is defined as:

- 1: Active
- 2: Disabled

5. After all checks, if the logged-in user type is a **general user**, display:
"Your role does not have access to this system."

If the user is an **administrator**, allow successful login and display:

"{Account.LastName} {Account.FirstName}, welcome."

The account type (Account.AccountType) is defined as:

- 1: General User
- 2: Administrator
- 3: Customer Service

Also, display the last successful login date and time. If no record exists, display:
"No successful login record found."

6. Clicking the **Logout** button will log out the current user and display:

"You have successfully logged out."

Then redirect back to the login page.

Register

1. Allow new account registration with data fields that must comply with the Account table schema.
2. AccountStatus and AccountType are both set to default value 1.
3. If the username already exists, reject the registration and display:
"Username already exists."

(2) Vehicle Query

- All administrators who can log in to this system are authorized to operate this function, which is used to manage the company's manufactured vehicle data.
- **Filter Functions (Search):**
 1. **Model:** The vehicle models produced by the company are **ZEVI** and **ZEVI**.
Selecting "All" means no filtering by model.
 2. **Country of Manufacture:** Options are **TW** (Taiwan) and **US**. Selecting "All" means no filtering by country.
 3. **Manufacturing Date:** Leaving this blank means no filtering. If only a start date is entered, the end date is ignored, and vice versa. The date format must be restricted to yyyy-MM-dd.
 4. **Keyword:** Partial matches on the Account table's **FirstName** or **LastName** fields.
Changing the keyword should immediately update the query results with the current matching data.
- **Sorting Function:**
 1. All columns except the data sequence number are sortable.
- **After Clicking Query, the Following Data Will Be Displayed:**
 1. **Data Sequence Number:** Only a sequential index, unaffected by sorting.

2. **Model:** Clicking the model opens a new window showing detailed information about the electric vehicle. If another record is clicked while the detail window is still open, the detail window content should update instead of opening a new window.
3. **VIN (Vehicle Identification Number):** Clicking the VIN opens a new window showing the vehicle's previous owners, displayed as {LastName}{FirstName}. If another VIN is clicked while the owners' window is still open, the content should update instead of opening a new window.
4. **Current Owner:**
 - If there is a current owner, display as {LastName}{FirstName}. Clicking the owner's name opens a new window showing detailed account information (all fields except password and ID must be displayed).
 - If another owner is clicked while the account detail window is open, update the content instead of opening a new window.
 - If there is no current owner, display **NA**.
5. **Manufacture Date:** Format yyyy-MM-dd.
6. **Country of Manufacture.**
7. **Battery Capacity:** Unit in kWh.
8. **Motor Power:** Unit in kW.
9. **Created Data Time:** Format yyyy-MM-dd HH:mm.
10. **Battery Health:** Use 85% as the threshold—values 85% or above show in green; below 85% show in red. New vehicles show 100%. The calculation is based on the maximum estimated capacity from the latest charging record in the BatteryChargeRecord table.
11. **Reference Screen:** (Screen data is for reference only)

Model:

Country of Manufacture:

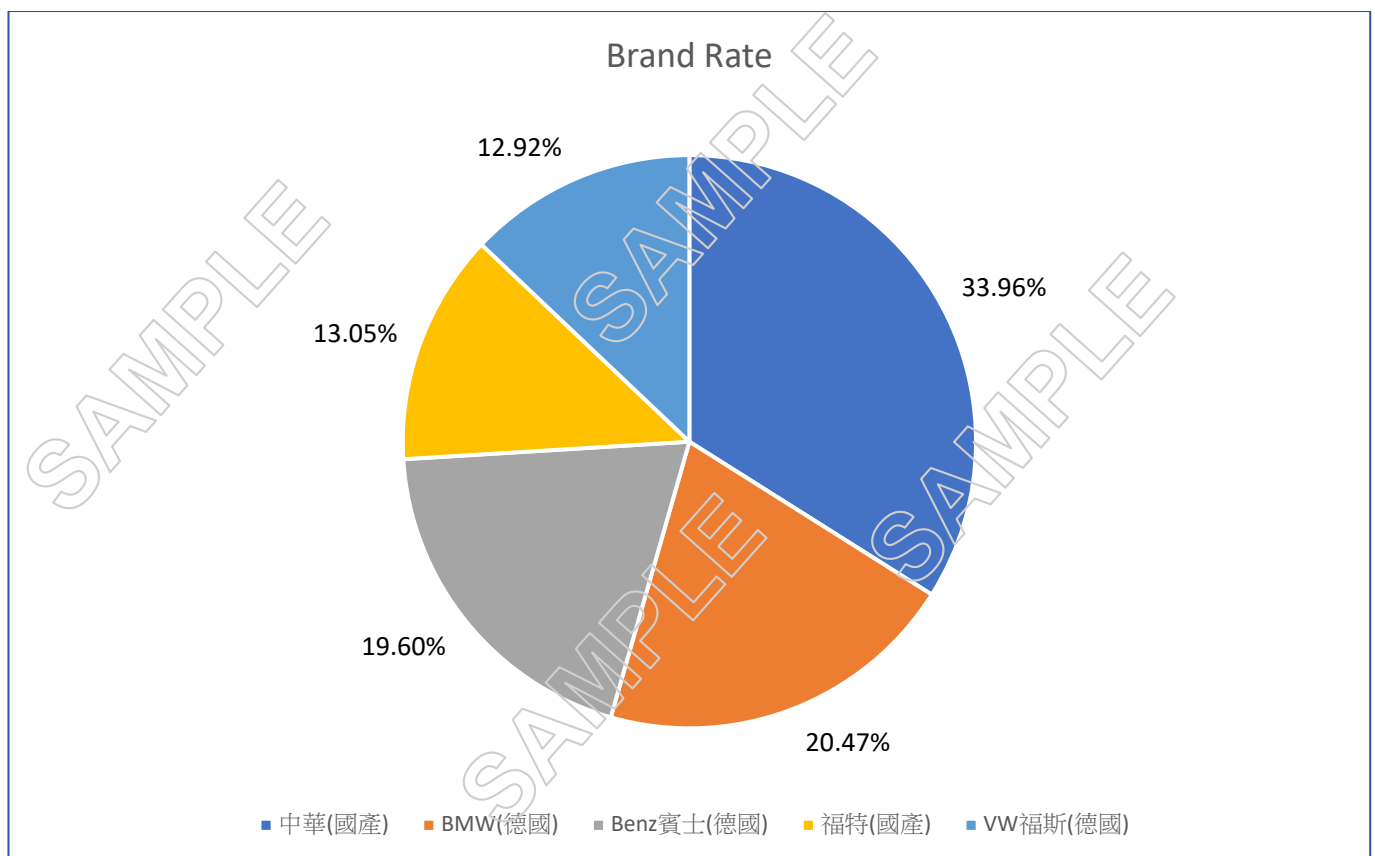
Manufacture Date: ~

Keyword:

	Model	VIN	Current Owner	Manufacture Date	Country of Manufacture	Battery Capacity	Motor Power	Created Date Time	Battery Health
1	ZEV1	YJMSLL74QCJC9OVCI	Morris Brett	2023-01-10	US	63 kWh	125 kW	2023-02-05 03:08	87.86%
2	ZEVX	GIO277XQJQU8V9Q84	NA	2021-08-31	TW	82 kWh	150 kW	2021-10-11 12:28	84.74%

(3) Display Pie Chart

- All user roles have read access to the **Brand** and **BrandModel** tables for statistical analysis. The pie chart should display the **top 5 vehicle brands** by market share. The percentages represent the proportion of these top 5 brands only, **not** the entire dataset. (Hardcoding data, reading fixed content, or using screenshots is **not allowed**.)
- Display the **Brand Rate** pie chart.
(Reference screen: The displayed data is for illustration purposes only.)



Description		Mark	Value
(1)	Instructions - Some system functions require you to add new columns. The added columns must start with New_ (e.g., New_MyColumn1), and the column names must be in English. Columns not following this rule will not be scored. - When submitting the SQL file, the filename must be Zyberion_ANM.bak (with the last two characters NM replaced by your position number). Failure to restore the database will result in a 20-point deduction. - When submitting code files, if the files are not correctly or completely uploaded to the FTP server, a 20-point deduction will be applied. Scoring checklist: - Is the database name Zyberion_ANM (with NM as your position number) correct? (1 point) - Is the SQL backup file named Zyberion_ANM.bak (with NM as your position number) correct? (1 point) - Is the solution file named ZCP.sln? (1 point) - Is the ReadMe.txt file name correct, and does its content include all user roles? (2 points, 1 point deducted for each error)	5	
(2)	User Interface Guidelines - The following elements must be persistently displayed: System Name, Brand Logo, System Time, and Current User's Name (4 points total, 1 point deducted for each missing or incorrect item) - Correctly display the current user's name in the format: {First Name}{single space}{Last Name} (1 point)	9	

Description		Mark	Value
	- Login/Logout status must be correctly displayed according to the current state (1 point) - Correct use of color palette (1 point) - Correct use of brand identification (logo, colors, etc.) (1 point) - Correct use of fonts (1 point)		
(3)	Login and Logout Functionality - Password input box defaults to hidden (masked) characters? (1 point) - Password is shown in plain text when mouse hovers over the “” icon, and reverts to masked when the mouse leaves? (3 points) - Display message "Account or password incorrect, error count: n times" based on conditions? (1 point) - If login errors exceed 3 attempts (not including 3), the account is set to disabled? (2 points) - Show account locked message when applicable? (1 point) - Display successful login message? (1 point) - Show last successful login date and time; if none, show "No successful login record found"? (2 points) - Correct implementation of logout functionality? (2 points, 1 point deducted for each error)	13	
(4)	Account Registration Functionality - Add new account data with all fields matching the Account table columns. (2 points) - Default values for AccountStatus and AccountType must both be set to 1. (1 point) - If the account already exists, registration is rejected with the message: "Account already exists." (2 points)	5	

	Description	Mark	Value
(5)	Vehicle Query Functionality • Access Control: Only administrators can use this function to manage the company's manufactured vehicle data. (2 points) Filtering Functionality (Search) • Model: Vehicle models produced by the company are ZEV1 and ZEVX. Choosing "ALL" means no filtering. (2 points) • Country of Manufacture: Options are TW (Taiwan) / US (United States). Choosing "ALL" means no filtering. (2 points) • Manufacture Date: No input means no filtering. If only the start date is entered, no end date filtering is applied, and vice versa. Date format must be restricted to yyyy-MM-dd. (4 points) • Keyword: Partial match on the FirstName or LastName fields in the Account table. Changing the keyword must update the query results accordingly. (3 points) Sorting Functionality • All fields except the data serial number can be sorted. (3 points, deduct 0.5 points per error) Display Data • Data Serial Number: Shows only the data sequence and is not affected by sorting. (1 point)	41	

	Description	Mark	Value
	Model: Clicking on the model opens a new window showing detailed vehicle information. If another model is clicked while the detail window is open, the data in the window should update instead of opening a new window. (5 points, deduct 2 points if a new window opens instead of updating) VIN (Vehicle Identification Number): Clicking the VIN opens a new window displaying all previous owners in the format {LastName}{FirstName}. If a different VIN is clicked while this window is open, update the content instead of opening a new window. (6 points, deduct 2 points if a new window opens instead of updating) Current Owner: If there is a current owner, display {LastName}{FirstName}. Clicking the owner opens a new window showing the owner's account details (excluding password and ID; all other fields must be shown). If another owner is clicked and the detail window is still open, update the content instead of opening a new window. If no current owner, display "NA." (3 points) Manufacture Date:Format must be yyyy-MM-dd. (1 point) Country of Manufacture: (1 point) Battery Capacity: Unit must be kWh. (1 point, deduct 0.5 points if unit not shown) Motor Power: Unit must be kW. (1 point, deduct 0.5 points if unit not shown) Data Creation Time: Format must be yyyy-MM-dd HH:mm. (1 point) Battery Health: Threshold is 85%. Battery health $\geq 85\%$ displayed in green; below 85% displayed in red. New cars have 100% battery health.		

	Description	Mark	Value
	Calculation based on the maximum estimated capacity in the last record of the BatteryChargeRecord table. (5 points, deduct 2 points if colors not displayed correctly)		
(6)	Displaying Pie Chart • All Account Permissions Can Access Statistics: All account types have permission to view the statistics. (1 point) • Data Source and Ranking: Read the Brand and BrandModel tables to perform statistical analysis. Display the top 5 vehicle brands by market share. The proportions shown should be relative only to the top 5 brands combined, not the entire dataset. (Do not hardcode values, read from the database dynamically, and do not use screenshots or fixed content.) (5 points) • Correct Pie Chart Style: The pie chart must be displayed with correct visual style (e.g., colors, segments, labels). (2 points) • Correct Chart Title: The pie chart must display the proper title. (1 point) • Correct Display of Top 5 Vehicle Brands: The chart must correctly show the names of the top 5 vehicle brands. (3 points) • Correct Display of Proportions: The proportions (percentages) shown on the chart must be accurate based on the top 5 brands data. (5 points)	17	
(7)	Programming Development Quality • Complete Input Validation, Error Prevention, and User Assistance Messages: The system must include thorough input validation (to prevent	10	

Description		Mark	Value
	user errors), error handling, and helpful user prompt messages. (3 points; deduct 1 point for each missing or incorrect implementation) • Clear and Understandable Code Comments: The source code should contain clear and meaningful comments explaining the logic and important sections. (3 points) • Meaningful Variable Naming Conventions: Variable names should be meaningful and descriptive, not just default or meaningless names generated by the system or IDE. (2 points) • Meaningful Form Naming Conventions: Form names (UI forms, pages, components, etc.) should have meaningful and descriptive names rather than default or generic ones. (2 points)		
Total		100	